

Machine learning techniques for stock price prediction and graphic signal recognition

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ABSTRACT

Stock market analysis is extremely important for investors because knowing the future trend and grasping the changing characteristics of stock prices will decrease the risk of investing capital for profit. Thereupon, the prediction of stock prices and identifying the graphic signals of candlestick charts, which are two crucial tasks in stock price analysis, attract much attention from investors owing to the returns and risks that coexist in financial markets. To introduce a reliable approach for addressing these challenges, this paper proposes the modeling strategies based on machine learning (ML) techniques. A vector autoregression (VAR)-based rolling prediction model is proposed for forecasting stock prices, and a Gaussian feed-forward neural networks (GFNN)-based graphic signal identification method is introduced to recognize different types of stock price signals. The experimental results demonstrate better performance comparing with the state-of-the-art methods, and it can be successfully applied in real-world stock exchange strategies.

1. Introduction

Stock price analysis is most important for the intelligent decision of financial markets and it plays a key role in stock trading strategies. Accurately predicting stock prices and identifying graphical signals of candlestick charts is of great realistic significance, since they can grasp the trend patterns of stock prices and help investors make appropriate investment strategies. Effective trading activities can be then implemented to attain greater returns with less risk (Li et al., 2022). However, with the rapid development of calculation capacity and machine learning techniques, the prediction models become increasingly complex and harder to interpret, which makes investors feel uneasy to use. Besides, as Buchanan (2009) and Farmer and Foley (2009) pointed out that classical economic models need to be revisited in the face of risks. Zhou et al. (2023) also stated that one of the potential issues existing in the stock market today is the increasingly complex interactions that are easy to be neglected. Other than that, the motivation for knowing the exact values of future stock prices makes most research focus on stock price prediction while ignoring the analysis of graphical signal characteristics. Hence, there are great needs and fundamental

importance to seek a new tool for automatically identifying graphical signals and forecasting stock prices.

In the existing methods of stock price prediction, the autoregressive integrated moving average (ARIMA) (Ariyo et al., 2014; Babu and Reddy, 2015), gray relational analysis (GRA) (Hamzaçebi and Pekkaya, 2011), rough set (Pal and Kar, 2019), and others are usually applied in real-world applications. Also, due to the strong interpretability and generalization ability, the commonly-used methods such as linear regression, moving average, and Generalized AutoRegressive Conditional Heteroskedasticity (GARCH) are favorable for financial time-series prediction (Long et al., 2019). These traditional methods perform well in linear and stationary time series but do not conduct ideally on the non-linear and non-stationary sequence data. In addition to the traditional statistical methods, data mining (DM) and machine learning (ML) algorithms have carved their niches in various prediction problems, such as Crude-Oil-Production forecasting (Al-Qaness et al., 2022a; Al-qaness et al., 2022c), CO2 trapping estimation (Al-Qaness et al., 2022d), and wind power prediction (Al-qaness et al., 2022b). They have also shown great potential in the field of financial time series forecasting. For example, using the method of support vector machine (SVM), Cao and Tay (2001) and Ince and Trafalis (2007) implemented the prediction

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research on the daily exchange data of stock prices. The satisfactory results are achieved by their proposed methods. However, the SVM is sensitive to the selection of kernel functions and their parameters, which is determined manually and has a certain degree of subjectivity. Besides, compared with other most-used ML methods, the generalization ability of SVM is relatively weak too. Sadowsky (2022) used several ML methods, such as random forests (RF), bagging, SVM, and extremely randomized trees, to predict stock prices. Their experimental results demonstrate that the decision tree (DT)-based techniques like RF, bagging, and extremely randomized trees outperform other ML methods. Whereas, different from a single algorithm, their used methods are ensemble learning methods comprised of multiple decision trees, which undoubtedly increases the complexity. In another research, Gerritsen et al. (2020) performed a trading rule mining analysis for the profitability in Bitcoin markets, and the experimental findings reveal that their method outperformed traditional methods. Nevertheless, this method relies entirely on the hand-designed rules and cannot be transplanted in a wide range. Since White (1988) first used artificial neural network (ANN) to forecast the daily return rate of IBM ordinary stock, the usage of the neural network has become a hotspot in stock price prediction due to its unique non-assumable, noise-tolerant, and adaptive properties. Even with complicated relationships between variables, the ANN methods (Xu et al., 2022; Yang et al., 2021; Zhang and Lou, 2021) have attained promising prediction and identification effects of stock time-series variation. However, the development of neural networks and the tasks related to architecture design, network hyper-parameters, and the constructed training samples require solid care to achieve the desired generalization level. In particular, the aforementioned methods mainly focus on the prediction of stock prices, but do not analyze the stock graphic signals. For this reason, Chen et al. (2021) studied the changes in stock prices and proposed a RVFL-GMDH model to predict the turning points rather than the exact values of stock prices. Although the change trends of stock prices are analyzed, the graphic signal identification of stock prices is not involved in their work. Recently, one of the novel ML methods, named deep learning (DL), has also been applied in this field. To utilize the power of a deep convolutional neural network (CNN), Sezer and Ozbayoglu (2018) converted a 1-D financial time series into a 2-D image-like data representation and suggested a DL model called CNN-TA for stock market forecasting. Whereas, this method consumes a large amount of work, especially, when the dataset is large-scale, and it requires a high training cost. Alonso-Monsalve et al. (2020) introduced a deep CNN for the high-frequency trend prediction of cryptocurrency exchange rates. Their experimental results indicate that the deep CNNs outperform conventional methods and are suitable for price trend prediction on most cryptocurrencies. Nevertheless, the large size of deep CNN models limits their usage. Owing to technological advances, the recurrent neural network (RNN) architecture, such as long-short-term memory (LSTM) and gated recurrent unit (GRU) networks, has also been employed to forecast stock prices (Saud and Shakya, 2020; Touzani and Douzi, 2021; Xu et al., 2022). Similar to the issues mentioned earlier, although reasonable good results have been reported in the literature, the abovementioned methods primarily predict the precise values of stock closing price, while ignoring the internal relationship analysis of time series. Moreover, most deep neural networks are usually cumbersome, with large volumes, and hard to deploy in mobile applications. A large number of parameters that need to be trained by massive labeled data is also a problem for DL models (Ładyżyński and Grzegorzewski, 2013). Despite the limitation, previous studies have successfully demonstrated the potential of ML and ANN methods for addressing both large-scale and small-scale problems.

Alternatively, so far, there has been little attention to using ML and ANN methods for identifying graphical signals of stock prices. Some of existing research focuses on the graphical signal identification of stock prices using pattern matching methods. Based on the template and rule approaches, Fu et al. (2007) proposed two flexible time

series pattern-matching approaches for graphical signal identification of stock prices. Among them, the template-based approach provides a tool for time series pattern matching, while the rule-based approach offers further ability for describing the query patterns. However, this approach is strictly depending upon the manually designed template and rule, which requires expert knowledge and is hard to promote. By using a concept of so-called fuzzy geometric protoform, Ładyżyński and Grzegorzewski (2013) identified the patterns in graphical visualizations of stock price data. Whereas, the computation process of this method is complicated and it has a certain ambiguity. Therefore, on the grounds of reviewing relevant literature, this paper investigates novel machine learning techniques for predicting stock prices and identifying graphic signals of stock time series. A vector autoregression (VAR)-based rolling prediction model is proposed to forecast stock prices. The value of the latest period is predicted using the VAR first, and then the prediction results combined with the most recent actual value are used as the input for the rolling prediction of the next stage. Using the stock prices as the input and various graphical signal types as the output, a Gaussian feed-forward neural networks (GFNN) is proposed and trained to identify the graphic signal types of stock time series. Multiple experiments have been implemented and the experimental findings demonstrate the validity and feasibility of the proposed procedure. In short, the key contributions of this study can be recapitulated as follows.

- To detect the pattern or direction of stock price fluctuations, we have summarized eight basic graphical signals of the stock price candlestick chart. The characteristics of each graphical signal are analyzed and the relevant trading strategies for each graphical signal of the candlestick chart have been provided as well.
- By applying the internal relationship of different stock prices, a VAR-based rolling prediction model is proposed for stock price forecasting. Especially, considering the rule that the weight is big when near and small when far, a strategy of using the latest prediction results coupled with the most recent value is adopted for the input of the model.
- After collecting historical data and tagging graphic signals, we established the expert sample data for the detection of graphic signals. Then, a GFNN-based model is proposed to identify the graphic signal types of stock prices, thereby assisting the exchange strategy of the stock market.

The remaining writing is decorated as: Section 2 introduces the materials used in this paper and importantly discusses the methodology, including the related works and the proposed approach. Section 3 is dedicated to the experimental analysis, a series of experiments are implemented in this section and the effects are evaluated. Section 4 concludes this paper with a summary and points out the direction of further work.

2. Materials and methods

2.1. Materials

The practical stock price data are used as the materials in our research on graphic signal identification and prediction analysis of stock prices. By searching the historical data of stock trade recordings from Yahoo Finance (<https://finance.yahoo.com>), we randomly chose several stocks as the analysis objects. The stock code of 600728.SS for Pci-Suntek and the stock code of TSCOL in the FTSE-100 index constituent were selected in our analysis. The data for the last 3 years were extracted by the Yahoo search engine. That is, the historical data of the Pci-Suntek and TSCOL stock prices from January 26, 2019 to January 26, 2022 were extracted from the Yahoo Finance website, respectively. Moreover, the forex trading data such as USDJPY and XAUUSD, which include the prices of open, close, high, and low, were also chosen as the materials in our analysis, and the sample data of the latest 2 years were extracted to perform the comparative experiments.

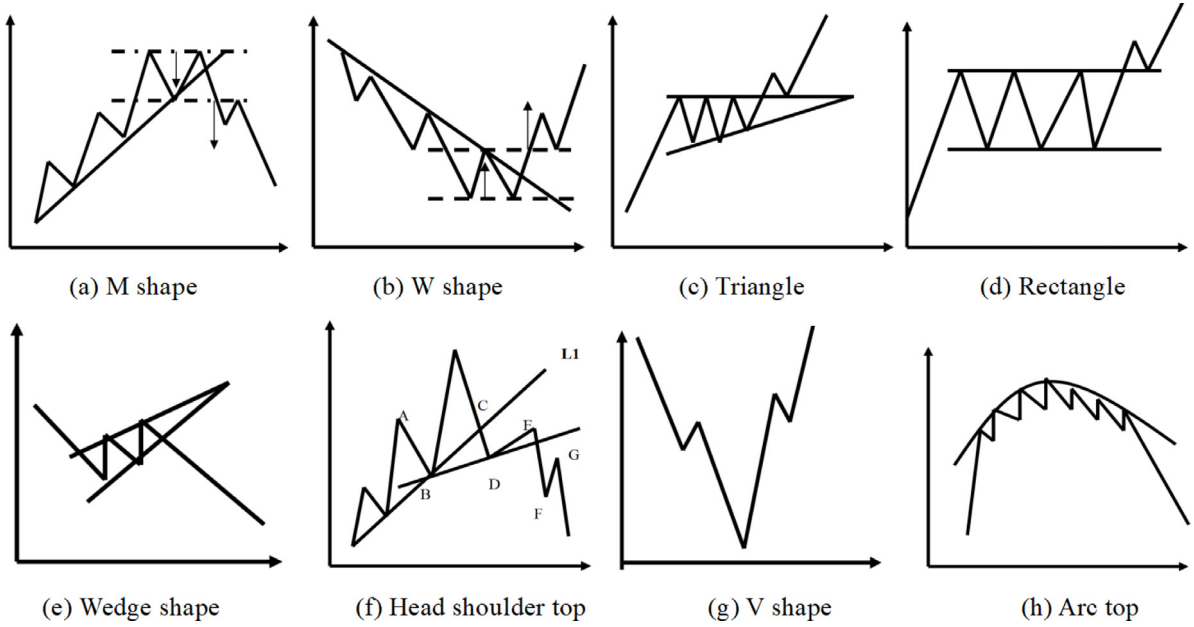


Fig. 1. The typical trend shapes of stock price graphical signals.

2.2. Methodology

2.2.1. Background and related works

(1) Typical graphical signals

To the best of our knowledge, investors want to know the future trend of stock price movements, such as rising or falling, and thus determine the trading strategies including long-term, short-term, buying, holding, selling, and so on. Therefore, for the purpose of predicting the pattern or direction of stock price fluctuations, the identification of candlestick chart signals in quantitative trading is of fundamental importance. Graphical signal identification is an essential part of technical analysis. It infers that the existing market trend will continue or reverse by analyzing various price patterns formed during the movement of stock time series. In general, there are eight typical graphical signals of the candlestick chart in the analysis of stock prices. Fig. 1 portrays the sample shapes of these graphical signals, and the detailed descriptions are presented below.

M-shape: also known as “double top” or “M” head, is one of the more common inversion forms in the candlestick chart. Once the M-shape is formed, the decline of the stock price is almost certain.

W-shape: is also called W-bottom or double-bottom. It usually appears at the bottom and is a typical form of a bottom reversal. This pattern must appear after the price drops from a high to a low. If the W-shape pattern appears in other parts, it needs to be operated carefully to avoid undue losses.

Triangle: it refers to the stock price moving up and down within a certain range after a period of rapid changes, and then continuing the previous trend at an appropriate time. It can be generally divided into three types, including symmetrical triangle, rising triangle, and falling triangle.

Rectangle: the rectangle, also known as a box shape, is a typical continuous consolidation pattern in which stock prices move sideways in two balanced straight lines.

Wedge shape: the wedge shape can be divided into ascending wedge and descending one, where the descending wedge generally forms a continuous downward trend while the rising wedge generates a constant upward trend.

Head shoulder top: the shape of the head shoulder top consists of the left shoulder, head, right shoulder, and neckline. Once the head and shoulder pattern is formed, it indicates that the top has been built. The

stock price will leave the previous upward trend and turn downward to enter a long-term downtrend.

V shape: V-shape is a reversal pattern that is difficult to predict. It often appears in violent market fluctuations without an obvious formation process. Indeed, the V shape is an out-of-control form and should be used with special care, in conjunction with basic information.

Arc top: the shape of the arc top is somewhat similar to the head and shoulder top but there is no obvious distinction between the head and shoulders. The arc top shape is mostly explosive, with a rapid rise and fall and a short duration. After this shape is determined, we should follow the trend and implement the trading strategy accordingly.

(2) Vector regression model

To our knowledge, vector regression is a typical machine learning model that has been popularly used for forecasting problems. The multilinear vector regression model is first introduced by Christopher Sims in 1980 and it is a generalization of the autoregressive moving average (ARMA) models formulated by Box and Jenkins (Guefano et al., 2021). The vector regression model can be used to model more than one variable jointly and it belongs to a multivariate regression model (Zhang and Hu, 2021).

Suppose that a multivariate time series y_t is input, the prediction of the vector regression model can be mathematically expressed as

$$y_{t+1} = c + A_1 y_t + A_2 y_{t-1} + \dots + A_p y_{t-p+1} + e_{t+1} \quad (1)$$

where c is an $n \times 1$ constant vector, A is an $n \times n$ matrix, and e implies the $n \times 1$ error vector. To keep it simple, for a y_t with two variables and one lag order, the formula of predicting y_t using the vector regressive model can be defined in

$$\begin{bmatrix} y_{1,t+1} \\ y_{2,t+1} \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} + \begin{bmatrix} A_{1,1} & A_{1,2} \\ A_{2,1} & A_{2,2} \end{bmatrix} \begin{bmatrix} y_{1,t} \\ y_{2,t} \end{bmatrix} + \begin{bmatrix} \epsilon_{1,t+1} \\ \epsilon_{2,t+1} \end{bmatrix} \quad (2)$$

Alternatively, it can also be written using the following equations:

$$y_{1,t+1} = c_1 + A_{1,1} y_{1,t} + A_{1,2} y_{2,t} + e_{1,t+1} \quad (3)$$

$$y_{2,t+1} = c_2 + A_{2,1} y_{1,t} + A_{2,2} y_{2,t} + e_{2,t+1} \quad (4)$$

where $y_{1,t}$ and $y_{2,t}$ represent the different variables included in the sequence data y_t , respectively. In brief, a vector regression model is known to show impressive performance in most cases of multivariate prediction problems, and it has shown a significant influence on a variety of prediction algorithms.

(3) Artificial Neural Networks

Artificial neural networks (ANNs) (Hastie et al., 2009; Schalkoff, 1997; Moon et al., 2018) are designed based on biological computational models that are formed by many interconnected neurons with coefficients composing the neural structure. The ANNs have become powerful tools in estimating and modeling various linear and non-linear systems. In general, there are many types of ANNs like feed-forward neural networks, Kohonen self-organizing networks, recurrent networks, and others. Among them, the feed-forward neural network, which is one of the most popularly used neural networks for forecasting tasks (Amber et al., 2015; Jovanović et al., 2015), consists of multiple layers such as the input, hidden, and output layers. The output of the neurons belonging to the hidden layer is determined by

$$y_j = \phi\left(\sum_{i=1}^m w_{ij}x_i + w_{j0}\right) \quad (5)$$

where x_i denotes the input neuron, w_{ij} is the weight between the i th neuron of the input layer and the j th neuron of the hidden layer, and ϕ signifies the activation function. The result of the output layer neuron is represented in a similar manner. The recurrent network architecture, such as the long-short-term memory (LSTM) and gated recurrent unit (GRU) networks (Shi et al., 2015; Cho et al., 2014), have become popular methods thanks to technological advances and successful blog posts by Karpathy and Olah. Also, combining aspects of RNN and CNN, the temporal convolutional network, or simply TCN (Bai et al., 2018), has emerged for time-series data prediction. More information about the basics and theoretical backgrounds of network architecture can be viewed from the study of Torres et al. (2021).

2.2.2. Proposed approach

(1) VAR-based rolling prediction model

Vector autoregression (VAR) integrates multiple variables in a unified way and jointly uses the information of these variables. It covers more information than a single time-series analysis method. Thereupon, the VAR can deliver more outstanding performance when it is used for forecasting problems, particularly for the prediction of stock time series containing multiple prices like open, close, high, and low. Assuming 2 variables and the lag period of p , the functions of VAR can be expressed as follows.

$$X_t = w_0 + \sum_{i=1}^p \alpha_i X_{t-i} + \sum_{i=1}^p \beta_i Y_{t-i} + \epsilon_t \quad (6)$$

$$Y_t = w'_0 + \sum_{i=1}^p \alpha'_i X_{t-i} + \sum_{i=1}^p \beta'_i Y_{t-i} + \epsilon'_t \quad (7)$$

where w_0 and w'_0 are the constants; $\alpha_i, \alpha'_i, \beta_i, \beta'_i$ ($i = 1, 2, \dots, p$) are the hyper-parameters; the means of the random disturbance terms ϵ_t and ϵ'_t are 0, and the variance is constant. The above-mentioned equations are described by a matrix as

$$\begin{bmatrix} X_t \\ Y_t \end{bmatrix} = \begin{bmatrix} w_0 \\ Y'_0 \end{bmatrix} + \begin{bmatrix} \alpha_1 \beta_1 \\ \alpha'_1 \beta'_1 \end{bmatrix} \begin{bmatrix} X_{t-1} \\ Y_{t-1} \end{bmatrix} + \dots + \begin{bmatrix} \alpha_p \beta_p \\ \alpha'_p \beta'_p \end{bmatrix} \begin{bmatrix} X_{t-p} \\ Y_{t-p} \end{bmatrix} + \begin{bmatrix} \epsilon_t \\ \epsilon'_t \end{bmatrix} \quad (8)$$

Further, the general form of the VAR can be obtained below.

$$X_t = w + \sum_{i=1}^p A_i X_{t-i} + \epsilon_t \quad (9)$$

In Eq. (9), w is the constant vector, A_i is the parameter matrix, X_{t-i} is the i th order lag of X_t , and ϵ_t is the error term. The rolling prediction method is utilized in our scheme, and particularly, considering the rule that the weight is big when near and small when far, the latest prediction results combined with the most recent value are used as the input for the prediction of the next stage. In summary, Algorithm 1 depicts the detailed procedure of the VAR-based rolling prediction model.

Algorithm 1: the VAR-based rolling prediction model.

Input: Time series of stock price $D = \{X_1, X_2, \dots, X_n\}$ where $X_t \in R^n, t = 1, 2, \dots, n$

- 1 **Begin**
- 2 Standardize the sample data using the scale function $\hat{X}_t = (X_t - \bar{X})/S$, where $\bar{X}$ denotes the mean and S implies the standard deviation.
- 3 **for** $i = 1; i \leq \text{pred_num}; i++$ **do**
- 4 Calculate the optimum lag order p^* with the formula of $p^* = \arg \min(AIC(p))$, where $AIC(\cdot)$ denotes the Akaike information criterion.
- 5 Using the VAR method with the optimum lag order p^* to predict the next stock prices, as written by $\text{Pred}_{t+1} = \text{VAR}(X_t, p^*)$. Note that the input variables consist of different prices (Here is 4.).
- 6 Combining the prediction results and the latest actual value as the input for the rolling prediction.
 $\text{Input}(X_{t+1}) = w \times \text{Pred}_{t+1} + (1 - w) \times \frac{X_t - \bar{X}}{S}$, where w is set to 0.5.
- 7 **end for**
- 8 Get the results of the VAR-based rolling prediction. //pred_num refers to the number of forecast periods.

Output:

- 9 The final sequence results of the rolling prediction $\{Y_1, Y_2, \dots, Y_n\}$.

(2) GFNN-based signal identification model

The primary types of neural networks include feed-forward, self-organizing and random networks. Among them, the radial basis function network in feed-forward networks is depending upon the theoretical foundation of nonlinear transformation followed by a linear conversion, and has the ability to map low-dimensional space to high-dimensional space. It has the advantages of the global and best approximation. Whilst, the training procedure is fast, easy to execute, and does not fall into local optimization, which makes it more widely used in practice.

The problem we want to solve can be described as follows: Given a set $\{x_i \in R^m | i = 1, 2, \dots, N\}$ containing N distinct points and a corresponding set $\{d_i \in R^1 | i = 1, 2, \dots, N\}$ with N real numbers, a function $F : R^m \rightarrow R$ needs to be found to satisfy the following Eq. (10).

$$F(x_i) = d_i, i = 1, 2, \dots, N \quad (10)$$

Here, the radial basis function is used to determine the form of the function $F(\cdot)$, as expressed by

$$F(x) = \sum_{i=1}^N w_i R(\|x - x_i\|) \quad (11)$$

In Eq. (11), w_i is the weight vector, $\|\cdot\|$ indicates the Euclidean norm, $\{R(\|x - x_i\|) | i = 1, 2, \dots, N\}$ is a set of N arbitrary functions, and x_i is the center of the radial basis function. As a widely used function in various fields, the Gaussian function can accelerate the convergence speed and avoid local optimum issues. Thus, the Gaussian function is used as the radial basis function in the neural networks, and the function $R(\cdot)$ can be defined in Eq. (12).

$$R(\|dist\|) = e^{-\|dist\|^2} = \exp\left(-\frac{1}{2\sigma^2} \|x - x_i\|^2\right) \quad (12)$$

As a consequence, the output result of the proposed networks is the linear weighted sum of the hidden layer outputs, as expressed in Eq. (13). The neuron unit and the architecture of the neuron network are depicted in Fig. 2.

$$y_j = F(x_p) = \sum_{i=1}^h w_{ij} \exp\left(-\frac{1}{2\sigma^2} \|x_p - x_i\|^2\right) \quad (13)$$

$j = 1, 2, \dots, n$

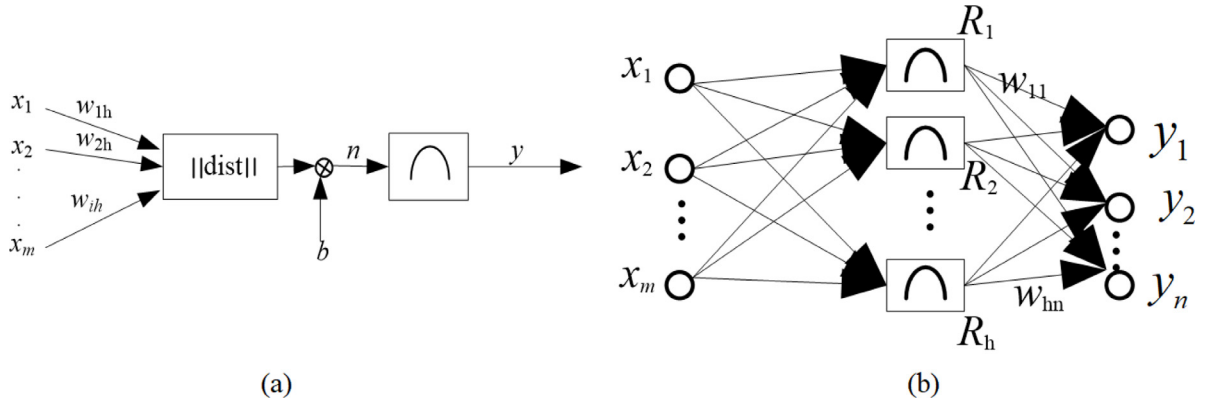


Fig. 2. The GFNN-based model (a) neuron unit; (b) the architecture of neuron network.

Table 1

The predicted results on the 600728.SS datasets.

	2022/1/20				2022/1/21				2022/1/24				2022/1/25			
	open	high	low	close	open	high	low	close	open	high	low	close	open	high	low	close
Actual data	9.19	9.3	9.01	9.23	9.25	9.38	9	9.07	8.94	9.1	8.85	8.94	8.95	9.01	8.41	8.46
GRU (Cho et al., 2014)	7.95	8.41	7.84	8.89	7.52	7.71	7.47	7.91	7.37	7.42	7.11	7.32	7.19	7.28	6.83	6.81
LSTM (Shi et al., 2015)	9.02	9.01	8.54	8.93	9.09	9.10	7.43	8.72	8.00	10.64	7.07	8.85	8.28	12	6.90	8.22
TCN (Bai et al., 2018)	8.79	8.83	8.44	8.73	8.54	8.78	8.49	8.45	8.73	9.14	8.50	8.67	9.63	9.62	8.75	9.08
This study	9.04	9.18	8.83	8.97	9.19	9.32	9.01	9.11	9.24	9.38	9.07	9.16	9.07	9.21	8.87	9.01

Table 2

The predicted results on the TSCO.L datasets.

	2022/1/20				2022/1/21				2022/1/24				2022/1/25			
	open	high	low	close	open	high	low	close	open	high	low	close	open	high	low	close
Actual data	289	289	286	287	285	288	285	288	287	291	287	288	289	292	286	291
GRU (Cho et al., 2014)	286	287	283	286	284	287	282	284	283	289	281	282	282	290	281	281
LSTM (Shi et al., 2015)	284	288	282	285	284	287	281	283	283	286	281	282	283	285	280	281
TCN (Bai et al., 2018)	290	291	284	287	294	290	285	287	289	290	284	289	290	290	285	290
This study	314	335	350	313	295	320	328	295	288	306	327	290	309	327	343	309

where y_j denotes the actual output of the j th node of the neuron network, h is the number of nodes in the hidden layer, and $\mathbf{x}_p = (x_1^p, x_2^p, \dots, x_m^p)^T$ means the p th input sample ($p = 1, 2, \dots, P$, P is the total number of samples.). On the grounds of this, the training samples of stock price time series are constructed first. The output is the graphical signal types of stock price candlestick charts, such as M-shape, W-shape, Triangle, Rectangle, among others, as mentioned in Section 2.2.1. The input is the transposition of the various stock prices, including open, high, low, and close prices. The time series window is set as 24. Since the output layer of the Gaussian feed-forward neural networks (GFNN) adjusts the weights using a linear optimization strategy, the learning speed is relatively faster and it is less risky to fall into local optimization. Besides, the input vector is directly mapped to the hidden layer space (no weighted connection or the weight is 1.), and once the expert sample is established, the mapping relationship can be learned, which makes the training procedure of the networks easy to execute. Thus, using the constructed sample data, we performed the GFNN-based signal prediction and identified the graphic signals of stock prices, thereby assisting the exchange strategy.

3. Experimental analysis

3.1. Experiment setup

We implement a series of experiments to validate the efficacy of the proposed procedure. Except for some classification algorithms executed by Modeler software, the primary algorithms are conducted using the R and PyCharm (Python 3.6), where frequently-used libraries like Tensorflow and Scikit-Learn (<https://scikit-learn.org/stable/>) are

employed for algorithm operation. The experimental hardware environment includes Intel® Core™ i7-8750 CPU @ 2.20 GHz, 16 GB memory, and NVIDIA GeForce GTX 1060 graphics card, which are used for model training and testing.

3.2. Results and evaluations

As mentioned in Section 2.1, we have extracted the historical data of stock prices including the Pci-Suntek (stock code: 600728.SS) and the FTSE-100 index constituent stock of Tesco (stock code: TSCO.L) for the prediction analysis. The historical data of stock prices from January 28, 2019 to January 25, 2022 were extracted. In addition to preserving some original data as the test set for assessing models, the ratio of the sample data randomly assigned to the validation set to those in the training set was 0.33. The former data were used to train the models while the latest ones were applied as the prediction sample. Similarly, for the forex trading data like USDJPY and XAUUSD, we have taken the historical data from 2020-12-21 22:00:00 to 2021-11-15 15:00:00, and the latest four periods of data are masked to validate models. Using the method proposed in Section 2.2.2, we performed the price prediction for these four stocks and the results are presented in Tables 1–4, respectively. Fig. 3 depicts the test performance of the proposed method on the 600728.SS dataset. Whilst, the 3 advanced methods including TCN, GRU, and LSTM were chosen to compare models. Kingma and Ba (2014) was used as the training optimizer of the models, with a 64 mini-batch size, 50 epochs, and a 1×10^{-3} learning rate.

Considering the measurement of model efficiency, we investigate the performance using different metrics like the accuracy (Acc), the

Table 3

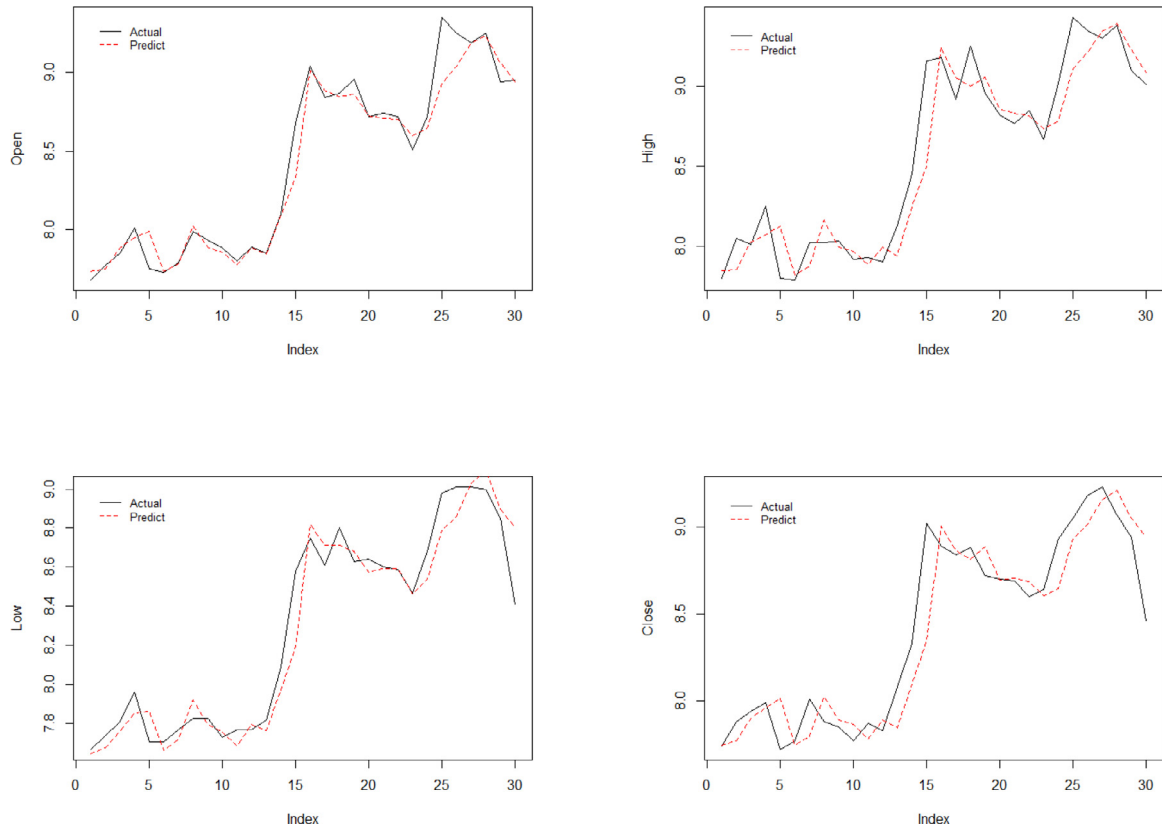
The predicted results on the USDJPY datasets.

	2021/11/15 13:30:00				2021/11/15 14:00:00				2021/11/15 14:30:00				2021/11/15 15:00:00			
	open	high	low	close	open	high	low	close	open	high	low	close	open	high	low	close
Actual data	113	113	113	113	113	113	113	113	113	114	113	113	113	114	113	113
GRU (Cho et al., 2014)	112	112	112	112	72	103	103	78	78	93	93	60	77	91	91	51
LSTM (Shi et al., 2015)	112	112	112	112	112	112	112	111	112	112	112	1111	111	111	111	112
TCN (Bai et al., 2018)	114	113	112	115	114	113	112	115	113	112	112	116	113	112	111	115
This study	113	113	113	113	113	113	113	113	113	113	113	113	113	114	113	113

Table 4

The predicted results on the XAUUSD datasets.

	2021/11/15 13:30:00				2021/11/15 14:00:00				2021/11/15 14:30:00				2021/11/15 15:00:00			
	open	high	low	close	open	high	low	close	open	high	low	close	open	high	low	close
Actual data	1867	1867	1862	1864	1864	1865	1861	1863	1863	1864	1859	1860	1860	1862	1858	1861
GRU (Cho et al., 2014)	1846	1865	1845	1844	1856	1956	1231	1665	1798	1839	1798	1852	1759	1897	1648	1869
LSTM (Shi et al., 2015)	1844	1923	1840	1846	2003	2022	1615	1933	2140	2163	1402	1704	2226	2247	1125	1697
TCN (Bai et al., 2018)	1864	1876	1865	1860	1866	1877	1870	1856	1867	1878	1871	1858	1868	1878	1871	1857
This study	1867	1869	1864	1867	1864	1867	1862	1865	1863	1865	1861	1863	1860	1863	1858	1860

**Fig. 3.** The validation performance of the proposed method on the 600728.SS dataset.

mean squared error (MSE), and the explained variance (E_{var}), which are correspondingly calculated in Eqs. (14)–(16).

$$Acc = 1 - MAPE = 1 - \frac{1}{n} \sum_{i=1}^n \frac{|y_i - \hat{y}_i|}{y_i} \quad (14)$$

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (15)$$

$$E_{var} = 1 - var(y_i - \hat{y}_i) / var(y_i) \quad (16)$$

where y_i is the observed value, $\hat{y}_i$ is the predicted value, $MAPE$ denotes the mean absolute percentage error between the observed

value and predicted value, and $var(\cdot)$ signifies the variance function. The indicator Acc reflects the prediction performance of the models, MSE signifies the deviation with the mean value of the squared error between the observed value and the predicted value, and the E_{var} indicates the explanatory power of the models. For both the Acc and E_{var} , the best possible value is equal to 1, while the bigger value is worse for the MSE indicator. Therefore, by this means, the performance measurements of the prediction results for different methods are summarized in Table 5, and the statistics of the Acc , MSE , and E_{var} are included in the table, respectively.

As seen in Table 5, the proposed approach has attained a competitive performance compared with other influential methods on the

Table 5The test accuracy of different methods (*Acc*: Accuracy).

Datasets	TCN (Bai et al., 2018)			GRU (Cho et al., 2014)			LSTM (Shi et al., 2015)			This study		
	Acc	MSE	E_{var}	Acc	MSE	E_{var}	Acc	MSE	E_{var}	Acc	MSE	E_{var}
600728.SS	0.9479	0.2525	-2.27	0.8393	2.2207	-1.14	0.9065	1.3296	-17.16	0.9771	0.0612	0.2302
TSCO.L	0.9936	7.3125	-0.726	0.9869	20	-0.439	0.9841	26.43	-0.3626	0.9041	1097	-80.93
USDJPY	0.9889	2.3751	-20.57	0.7944	883.62	-3135	0.9873	2.437	-2.3928	0.9994	0.0625	0.4642
XAUUSD	0.9959	78.625	-7.055	0.9496	31863	-3988	0.8801	84971	-13068	0.9993	2.625	0.7908
Average	0.9816	22.141	-7.655	0.8926	8192.2	-1781	0.9395	21250	-3271.9	0.9699	274.9	-19.86

Table 6

The partial tagged labels of USDJPY price data.

Time	Open	High	Low	Close	Tagged labels
2021-02-26 13:00:00+0000	106.242	106.347	106.217	106.339	Ascending triangle
2021-02-26 13:30:00+0000	106.338	106.338	106.246	106.305	Ascending triangle
2021-02-26 14:00:00+0000	106.304	106.396	106.282	106.382	Ascending triangle
2021-02-26 14:30:00+0000	106.381	106.449	106.371	106.448	Ascending triangle
2021-02-26 15:00:00+0000	106.446	106.512	106.382	106.502	Ascending triangle
2021-10-29 20:30:00+0000	114.001	114.015	113.909	113.978	Head shoulder top
2021-10-31 21:00:00+0000	114.015	114.064	114.003	114.051	Head shoulder top
2021-10-31 21:30:00+0000	114.052	114.098	114.052	114.089	Head shoulder top
2021-10-31 22:00:00+0000	114.073	114.152	114.056	114.146	Head shoulder top
2021-10-31 22:30:00+0000	114.135	114.152	114.074	114.14	Head shoulder top

experimental datasets, even if the optimum recurrent and convolutional neural networks are adopted. The proposed approach attains an average *Accuracy* of 96.99%. It outperforms the GRU and LSTM methods and has a comparative performance with the TCN, while the number of datasets that the proposed approach has achieved a higher accuracy is more than that of the TCN method. In brief, the proposed approach shows superior performance to the other comparative methods. Additionally, it is essential to emphasize that the time consumption of the proposed approach is 15.56 s, which is much less than the average time of 11.46 min consumed by the comparison methods. From the experimental results, it can be concluded that the proposed approach has achieved significant advantages in both accuracy and efficiency. The crucial explanation for the substantial performance of the proposed approach is that the VAR-based rolling prediction model integrates multiple variables in a unified way and explicitly models the interaction impacts within different stock prices of history. Particularly, the prediction results combined with the latest actual value are used as the input for the prediction of the next stage, which fully utilizes the rule that the recent the larger effect. Conversely, other methods only consider the stock price time series itself, and do not jointly use the information of different prices. Thereupon, compared with other state-of-the-art methods, the proposed approach has attained satisfactory effectiveness in the prediction of stock prices.

Furthermore, the forex trading data such as the USDJPY and XAUUSD were selected to implement the test of graphic signal identification. As shown in Table 6, the graphic signals of partial samples were manually tagged first. Then, the different price data including the open, high, low, and close prices were transposed and the time window was set as 24, thereby building the input samples. In other terms, the transposed stock price data were selected as the input while the tagged signal labels were used as the output. Table 7 presents the form of input data and the partial samples are included in it. After constructing the sample data, we performed the graphic signal identification of stock prices using the GFNN introduced in Section 2.2.2. The hyperparameters of the networks are assigned as follows: The number of network layers is set to 3, the number of hidden layer neurons is 8, the maximum epoch is 500, and the minimum error is 0.01. Fig. 4 portrays the graphic signal examples of stock prices.

Taking the accurate and wrong identification into account, we use the *Recall* (*Re*) metrics to evaluate the performance of the model. It is worth mentioning that the extended calculation for multi-classification tasks of this metrics is using micro averaging. The formula is presented as follows.

$$Recall = \frac{TP_1 + TP_2 + \dots + TP_C}{TP_1 + TP_2 + \dots + TP_C + FN_1 + FN_2 + \dots + FN_C} \quad (17)$$

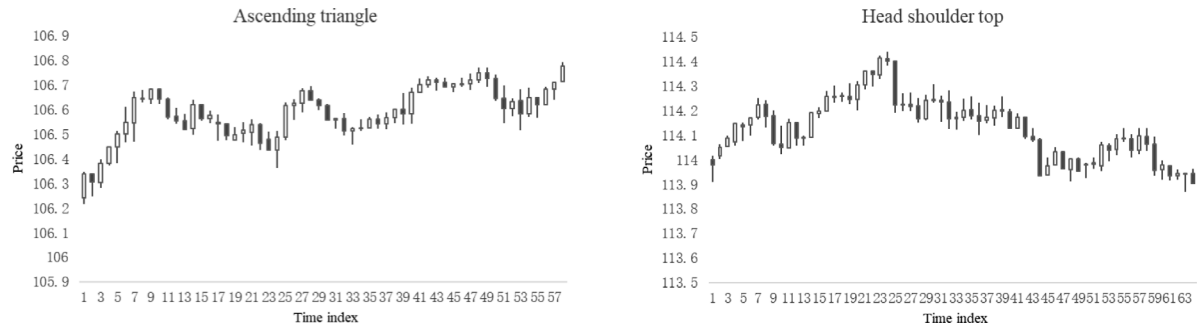
where C indicates the number of graphic signal classifications, TP_c and FN_c denote the true positive and false negative, which are the number of accurately and incorrectly identified graphic signals for a certain category c ($c=1,2,\dots,C$). Moreover, to further verify the effect of the model, three well-known algorithms including SVM, K-nearest neighbors (KNN), and decision tree are selected for the comparative experiments. Table 8 summarizes the micro recall rate of the results identified by different methods on the experimental datasets, and the representative identification samples of the proposed method are presented in Table 9.

It can be seen from Table 8 that the proposed method has delivered a satisfactory result and outperforms other state-of-the-art methods. The recall rate of the GFNN has separately reached 67.06% and 64.17% on the USDJPY and XAUUSD datasets, which are the top performance of all the algorithms. Besides, as seen in Table 9, the GFNN-based signal identification method has recognized most of the graphic signals for the stock price samples, and the predicted graphic signal labels are generally consistent with the actual manually-identified labels. For example, the samples with the manually-identified labels of “Ascending wedge” and “Descending wedge” have been successfully identified by the proposed method. The “V-shape” was recognized as the “Inverted V-shape”. Although individual categories were recognized incorrectly, the identified shape characteristics are compatible with the real graphic signal categories of stock prices. The main reason behind this is that the nonlinear transformation and Gaussian radial basis function makes the GFNN noise-tolerant, adaptive properties, and fast convergence, thereby obtaining strong approximation capability and grasping the useful information in the constructed sample data. By contrast, constrained by the approximation capability and data noises, the other methods do not achieve the desired effects. More than that, we have also conducted a performance investigation for our method compared to the findings reported in the previous literature. Chen et al. (2021) extracted the stock price data of 600728.SS and TSCO.L as the analysis objects and predicted the turning point signals of these two stocks. However, this research does not forecast the extract values of the stock prices, which is inconvenient for the stock exchange operation. In addition, Fu et al. (2007) performed a technical analysis of stock prices and summarized five typical graphic signals of stock time series. They proposed a template- and rule-based approach to achieve the goal of pattern recognition for stock prices. Though some graphic signals of stock prices can be identified, this approach is cumbersome, time-consuming, lags in prediction, and is not easy to be extended to a wide range. Also, there are relatively few summarized graphic signals. In comparison, the proposed approach is more feasible for

Table 7

The transposed sample data (a form of input data).

Prices	X1	X2	X3	X4	X5	...	X21	X22	X23	X24	Tagged labels
Open	106.242	106.338	106.304	103.381	106.446	...	106.506	106.539	106.478	106.439	Ascending triangle
High	106.347	106.338	106.396	106.449	106.512	...	106.561	106.546	106.51	106.512	Ascending triangle
Low	106.217	106.246	106.282	106.371	106.382	...	106.453	106.435	106.436	106.362	Ascending triangle
Close	106.339	106.305	106.382	106.448	106.502	...	106.539	106.462	106.436	106.487	Ascending triangle
Open	114.001	114.015	114.052	114.073	114.135	...	114.304	114.36	114.344	114.412	Head shoulder top
High	114.015	114.064	114.098	114.152	114.152	...	114.361	114.36	114.422	114.44	Head shoulder top
Low	113.909	114.003	114.052	114.056	114.074	...	114.284	114.298	114.33	114.383	Head shoulder top
Close	113.978	114.051	114.089	114.146	114.140	...	114.361	114.347	114.413	114.401	Head shoulder top

**Fig. 4.** The graphic signal examples of stock prices.**Table 8**

The identified recall rate of different methods.

Dataset	Methods	No. of samples	No. of labeled samples	No. of test samples	No. of correct samples	Recall rate
USDJPY	SVM	11,172	2041	932	576	61.80%
	KNN	11,172	2041	932	618	66.20%
	DT	11,172	2041	932	547	58.69%
	GFNN	11,172	2041	932	625	67.06%
XAUUSD	SVM	10,605	1683	374	184	49.20%
	KNN	10,605	1683	374	231	61.76%
	DT	10,605	1683	374	216	57.75%
	GFNN	10,605	1683	374	240	64.17%

Table 9

The representative samples of GFNN-based signal identification results.

Time	Open	High	Low	Close	Tagged label	Identified label
2021-01-15 09:30:00	1852.626	1856.935	1852.485	1855.152	Ascending wedge	Ascending wedge
2021-01-15 10:00:00	1855.106	1855.354	1853.096	1853.72	Ascending wedge	Ascending wedge
2021-01-15 10:30:00	1853.767	1855.812	1853.044	1853.898	Ascending wedge	Ascending wedge
2021-01-20 09:30:00	1856.353	1857.178	1854.354	1854.608	V-Shape	Inverted V-shape
2021-01-20 10:00:00	1854.616	1854.744	1851.701	1852.379	V-Shape	Inverted V-shape
2021-01-20 10:30:00	1852.368	1855.114	1852.273	1853.679	V-Shape	Inverted V-shape
2021-01-25 13:30:00	1867.062	1867.496	1863.033	1864.91	Descending wedge	Descending wedge
2021-01-25 14:00:00	1864.914	1866.232	1863.085	1864.277	Descending wedge	Descending wedge
2021-01-25 14:30:00	1864.332	1865.856	1854.925	1855.77	Descending wedge	Descending wedge

the prediction of stock prices and the graphic signal identification of candlestick charts. Based on the results of experimental analysis, it can be concluded that the proposed approach is effective in predicting stock prices and identifying graphic signals. It can also be transplanted to related domains, such as power load prediction, crude price analysis, and others.

4. Conclusions

The prediction of stock prices and the identification of graphical signals is crucial for the strategy of stock trading. It can help investors determine the buy or sell strategies for profit according to the knowledge of predicted stock prices and the identified graphical signal features. In this study, a VAR-based rolling prediction method was

proposed to forecast stock prices, and a GFNN-based graphic signal identification model was introduced to recognize the graphic signal classes of candlestick charts. The proposed prediction method achieves an average accuracy of 96.99% while consuming much less time. Also, the graphic signal recognition model reaches a minimum recall rate of 64.17% in practical business scenarios. The experimental findings show the feasibility and effectiveness of the proposed procedure, which has successfully predicted the prices and identified the graphics signals of stock time series. In future development, we wish to deploy the model on portable devices to automatically identify the graphics signals and predict the stock prices. Whilst, we would like to use it on more real-life applications like cryptocurrency market analysis, price trend prediction of Bitcoin and Ethereum, etc.

CRediT authorship contribution statement

Junde Chen: Methodology, Software, Writing – original draft.
Yuxin Wen: Help revise, Proofread, Supervision. **Y.A. Nanehkaran:** Experiments, Helping the revision of original paper, Proofread, English checking. **M.D. Suzaiddola:** Data curation, Helping the experiments and analyzed results. **Weirong Chen:** Project support, Software. **Defu Zhang:** Providing the support of the project, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

- Al-Qaness, M.A., Ewees, A.A., Abualigah, L., AlRassas, A.M., Thanh, H.V., Abd Elaziz, M., 2022a. Evaluating the applications of dendritic neuron model with metaheuristic optimization algorithms for crude-oil-production forecasting. *Entropy* 24 (11), 1674.
- Al-qaness, M.A., Ewees, A.A., Elaziz, M.A., Samak, A.H., 2022b. Wind power forecasting using optimized dendritic neural model based on seagull optimization algorithm and aquila optimizer. *Energies* 15 (24), 9261.
- Al-qaness, M.A., Ewees, A.A., Fan, H., AlRassas, A.M., Abd Elaziz, M., 2022c. Modified aquila optimizer for forecasting oil production. *Geo-Spatial Inf. Sci.* 1–17.
- Al-Qaness, M.A., Ewees, A.A., Thanh, H.V., AlRassas, A.M., Dahou, A., Elaziz, M.A., 2022d. Predicting CO₂ trapping in deep saline aquifers using optimized long short-term memory. *Environ. Sci. Pollut. Res.* 1–15.
- Alonso-Monsalve, S., Suárez-Cetrulo, A.L., Cervantes, A., Quintana, D., 2020. Convolution on neural networks for high-frequency trend prediction of cryptocurrency exchange rates using technical indicators. *Expert Syst. Appl.* 149, 113250.
- Amber, K., Aslam, M., Hussain, S., 2015. Electricity consumption forecasting models for administration buildings of the UK higher education sector. *Energy Build.* 90, 127–136.
- Ariyo, A.A., Adewumi, A.O., Ayo, C.K., 2014. Stock price prediction using the ARIMA model. In: 2014 UKSim-AMSS 16th International Conference on Computer Modelling and Simulation. IEEE, pp. 106–112.
- Babu, C.N., Reddy, B.E., 2015. Prediction of selected Indian stock using a partitioning–interpolation based ARIMA–GARCH model. *Appl. Comput. Inf.* 11 (2), 130–143.
- Bai, S., Kolter, J.Z., Koltun, V., 2018. An empirical evaluation of generic convolutional and recurrent networks for sequence modeling. *arXiv preprint arXiv:1803.01271*.
- Buchanan, M., 2009. Meltdown modelling: Could agent-based computer models prevent another financial crisis? *Nature* 460 (7256), 680–683.
- Cao, L., Tay, F.E., 2001. Financial forecasting using support vector machines. *Neural Comput. Appl.* 10 (2), 184–192.
- Chen, J., Yang, S., Zhang, D., Nanehkaran, Y.A., 2021. A turning point prediction method of stock price based on RVFL-GMDH and chaotic time series analysis. *Knowl. Inf. Syst.* 63 (10), 2693–2718.
- Cho, K., Van Merriënboer, B., Bahdanau, D., Bengio, Y., 2014. On the properties of neural machine translation: Encoder-decoder approaches. *arXiv preprint arXiv:1409.1259*.
- Farmer, J.D., Foley, D., 2009. The economy needs agent-based modelling. *Nature* 460 (7256), 685–686.
- Fu, T.-c., Chung, F.-l., Luk, R., Ng, C.-m., 2007. Stock time series pattern matching: Template-based vs. rule-based approaches. *Eng. Appl. Artif. Intell.* 20 (3), 347–364.
- Gerritsen, D.F., Bouri, E., Ramezanifar, E., Roubaud, D., 2020. The profitability of technical trading rules in the Bitcoin market. *Finance Res. Lett.* 34, 101263.
- Guefano, S., Tamba, J.G., Azong, T.E.W., Monkam, L., 2021. Methodology for forecasting electricity consumption by grey and vector autoregressive models. *MethodsX* 8, 101296.
- Hamzaçebi, C., Pekkaya, M., 2011. Determining of stock investments with grey relational analysis. *Expert Syst. Appl.* 38 (8), 9186–9195.
- Hastie, T., Tibshirani, R., Friedman, J.H., Friedman, J.H., 2009. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, Vol. 2. Springer.
- Ince, H., Trafalis, T.B., 2007. Kernel principal component analysis and support vector machines for stock price prediction. *Int. Trans.* 39 (6), 629–637.
- Jovanović, R.Ž., Sretenović, A.A., Živković, B.D., 2015. Ensemble of various neural networks for prediction of heating energy consumption. *Energy Build.* 94, 189–199.
- Kingma, D.P., Ba, J., 2014. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*.
- Ładyżyński, P., Grzegorzewski, P., 2013. Particle swarm intelligence tuning of fuzzy geometric protoforms for price patterns recognition and stock trading. *Expert Syst. Appl.* 40 (7), 2391–2397.
- Li, G., Zhang, A., Zhang, Q., Wu, D., Zhan, C., 2022. Pearson correlation coefficient-based performance enhancement of broad learning system for stock price prediction. *IEEE Trans. Circuits Syst. II* 69 (5), 2413–2417.
- Long, W., Lu, Z., Cui, L., 2019. Deep learning-based feature engineering for stock price movement prediction. *Knowl.-Based Syst.* 164, 163–173.
- Moon, J., Park, J., Hwang, E., Jun, S., 2018. Forecasting power consumption for higher educational institutions based on machine learning. *J. Supercomput.* 74 (8), 3778–3800.
- Pal, S.S., Kar, S., 2019. Time series forecasting for stock market prediction through data discretization by fuzzistics and rule generation by rough set theory. *Math. Comput. Simulation* 162, 18–30.
- Sadorsky, P., 2022. Forecasting solar stock prices using tree-based machine learning classification: How important are silver prices? *North Am. J. Econ. Finance* 101705.
- Saud, A.S., Shakya, S., 2020. Analysis of look back period for stock price prediction with RNN variants: A case study on banking sector of NEPSE. *Procedia Comput. Sci.* 167, 788–798.
- Schalkoff, R.J., 1997. *Artificial Neural Networks*. McGraw-Hill Higher Education.
- Sezer, O.B., Ozbayoglu, A.M., 2018. Algorithmic financial trading with deep convolutional neural networks: Time series to image conversion approach. *Appl. Soft Comput.* 70, 525–538.
- Shi, X., Chen, Z., Wang, H., Yeung, D.-Y., Wong, W.-K., Woo, W.-c., 2015. Convolutional LSTM network: A machine learning approach for precipitation nowcasting. *Adv. Neural Inf. Process. Syst.* 28.
- Torres, J.F., Hadjout, D., Sebaa, A., Martínez-Álvarez, F., Troncoso, A., 2021. Deep learning for time series forecasting: A survey. *Big Data* 9 (1), 3–21.
- Touzani, Y., Douzi, K., 2021. An LSTM and GRU based trading strategy adapted to the Moroccan market. *J. Big Data* 8 (1), 1–16.
- White, H., 1988. Economic prediction using neural networks: The case of IBM daily stock returns. In: *ICNN*, Vol. 2. pp. 451–458.
- Xu, H., Chai, L., Luo, Z., Li, S., 2022. Stock movement prediction via gated recurrent unit network based on reinforcement learning with incorporated attention mechanisms. *Neurocomputing* 467, 214–228.
- Yang, Z., Keung, J., Kabir, M.A., Yu, X., Tang, Y., Zhang, M., Feng, S., 2021. AComNN: Attention enhanced compound neural network for financial time-series forecasting with cross-regional features. *Appl. Soft Comput.* 111, 107649.
- Zhang, R., Hu, J., 2021. Production performance forecasting method based on multivariate time series and vector autoregressive machine learning model for waterflooding reservoirs. *Petrol. Explor. Dev.* 48 (1), 201–211.
- Zhang, D., Lou, S., 2021. The application research of neural network and BP algorithm in stock price pattern classification and prediction. *Future Gener. Comput. Syst.* 115, 872–879.
- Zhou, Y., Chen, Z., Liu, Z., 2023. Dynamic analysis and community recognition of stock price based on a complex network perspective. *Expert Syst. Appl.* 213, 118944.